§14. The koszul complex

Def. ① $r \in R$, $r \notin k$ is a non-zero divisor on U if $ru \neq 0$, $\forall\, 0 \neq u \in U$

(or U-regular)

② $f_1, \cdots, f_q \in R$ is U-regular sequence if

(1) $(f_1, \cdots, f_q)U \neq U$

(2) $\forall\, 1 \leq i \leq q$, f_i is a non-zero divisor on $U/(f_1, \cdots f_{i-1})U$.

- The koszul complex $K(\underline{f})$ on $f_1, \cdots, f_q$ ($\underline{f}$: a sequence $f_1, \cdots, f_q \in R$)

Construction:

$E = k\langle e_1, \cdots, e_q\rangle / \langle \{e_i^2 \mid 1 \leq i \leq q\}, \{e_ie_j + e_je_i \mid 1 \leq i < j \leq q\}\rangle$ $\wedge V$

$K(\underline{f})$: R-mod $R \otimes E$ with differential

$$d(e_{j_1} \wedge \cdots \wedge e_{j_i}) = \sum_{1 \leq p \leq i} (-1)^{p+1} f_{j_p} \cdot e_{j_1} \wedge \cdots \wedge \hat{e}_{j_p} \wedge \cdots \wedge e_{j_i}$$

(Verify: $d^2 = 0$)

Let $K(\underline{f}): 0 \to K_q \xrightarrow{d} \cdots K_1 \xrightarrow{d} K_0 \to 0$ where K_i is free R-mod with a basis $\{e_{j_1} \wedge \cdots \wedge e_{j_i} \mid 1 \leq j_1 < \cdots < j_i \leq q\}$ $\binom{p}{i}$

$K(\underline{f})$ is called koszul complex on $f_1, \cdots, f_q$.

- W f.g. R-mod, $K(\underline{f}; W) = K(\underline{f}) \otimes_R W$

Lemma 14.5 Let $\mathcal{F} = \{f_1, \cdots, f_{q-1}\}$, $f_q \in R$

$f = \{f_1, \cdots, f_q\}$

$\Rightarrow$ There is an exact seq of complexes

$$0 \to K(\mathcal{F}) \to K(f) \to K(\mathcal{F})[-1] \to 0$$

Pf. $K(f)_i = K(\mathcal{F})_i \oplus K(\mathcal{F})_{i-1} \wedge e_q$.

$e_{j_1} \wedge \cdots \wedge e_{j_i}$

$\square$

Rmk. $0 \to K(\mathcal{F}) \to K(f) \to K(\mathcal{F})[-1] \to 0$.

$\Rightarrow$ $H_{i-1}(K(\mathcal{F})) \xrightarrow{(-1)^{i+1} f_q} H_{i-1}(K(\mathcal{F}))$

$\shortparallel$

$H_i(K(\mathcal{F})[-1])$

Thm 14.7. W f.g R-mod $f=(f_1,\cdots,f_q)\in R$ of positive degree.

T.F.A.E

(1) $H_i(K(f;W))=0 \quad \forall i>0$
$H_0(K(f;W))=W/(f)W$

(2) $H_1(K(f;W))=0$.

(3) f is W-regular seq.

Pf. (1) $\Rightarrow$ (2) ✓

(3) $\Rightarrow$ (1) $\quad \to K_1\otimes W \xrightarrow{(f_1,\cdots,f_q)} K_0\otimes W \to 0$

$\Rightarrow H_0(K(f;W))=W/(f)W$.

By induction on q.

$q=1 \quad K(f_1;W): 0\to W\xrightarrow{f_1} W\to 0$

$\Rightarrow H_1=\{m\in W \mid f_1m=0\}=0$

Suppose it holds for $\underline{\bar{f}=\{f_1,\cdots,f_{q-1}\}}$

By Lemma 14.5

$$\begin{array}{ccccc} H_1(\bar{K}) & \to & H_1(K(f;W)) & \to & \underline{H_0(\bar{K})}=W/(\bar{f})W \\ \| & & & & \downarrow f_q \\ 0 & & & & H_0(\bar{K}) \\ & & & & \| \\ & & & & W/(\bar{f})W \end{array}$$

$\Rightarrow H_1(K(f;W))=0$.

$i>1 \quad 0=\underline{H_i(\bar{K})}\to H_i(K(f;W))\to \underline{H_{i-1}(\bar{K})}=0$

$\Rightarrow H_i(K(f;W))=0$.

(2) $\Rightarrow$ (3). Induction on q.

① $W\neq (f)W$

Assum $W/(f)W=0$.

$f_1,\cdots,f_q$ positive degree

$\xrightarrow{\text{Nakayama}} W=0$

$\Rightarrow W/(f)W\neq 0$.

② By lemma 14.5

$$H_1(\bar{K}) \xrightarrow{f_q} H_1(\bar{K}) \to \underline{H_1(K(f;W))} = 0$$

$\Rightarrow H_1(\bar{K}) = (f_q) H_1(\bar{K})$

$\Rightarrow H_1(\bar{K}) = 0$

Suppose $\underline{\bar{f} = (f_1, \cdots, f_{q-1}) \text{ is } W\text{-regular}}$

$$0 = \underline{H_1(K(f;W))} \to H_0(\bar{K}) \xrightarrow{f_q} H_0(\bar{K})$$

$$\begin{array}{cc} \| & \| \\ W/(\bar{f})W & W/(\bar{f})W \end{array}$$

$\Rightarrow f_q$ is non-zero divisor in $W/(\bar{f})W$

$\Rightarrow f$ is W-regular seq. $\square$

Cor 14.8. W f.g. graded R-mod, $f = (f_1, \cdots, f_q)$ $f_i \in m$, $f_1, \cdots, f_q$ homogeneous W-reg seq.

(1) Any permutation of $f_1, \cdots, f_q$ is W-reg

(2) If $u_1 f_1 + \cdots + u_q f_q = 0$ for $u_i \in W$, then $\forall i$, $u_i \in (f_1, \cdots, f_q) W$

(3) $k = S/(x_1, \cdots, x_n)$ as S-mod.
Koszul complex $\mathbb{K}(x_1, \cdots, x_n)$ is the m.g.f.r of k

$\Rightarrow b_i^S(k) = \binom{n}{i}$, $pd_S(k) = n$

$K_i = S(-i)^{\binom{n}{i}}$

Thm. V f.g graded S-mod

$$b_{i,p}^S(V) = \dim_k(\mathrm{Tor}_i^S(V, k)_p) = \dim_k H_i(\mathbb{K} \otimes V)_p$$

prop. $\mathrm{Tor}_S^n(S/I, k) \cong \mathrm{soc}(S/I) \overset{\Delta}{=} \{f \in S/I \mid mf = 0\}$

$S = k[x_1, \dots, x_n]$
$R = S/I$

§15. Finite projective dimension.

Thm. U f.g graded R-mod, then $pd_R(U) \le pd_R(k)$

pf. $\mathbb{F}_k$ m.g.f.r of k. Then

$$b_i^R(U) = \dim_k(\mathrm{Tor}_i^R(U,k)) = \dim_k(H_i(U \otimes \mathbb{F}_k))$$

$i > pd_R(k)$, $b_i^R(U) = 0$.

Hilbert's syzygy's Thm. U S-mod $pd_S(k) = n$

$pd_S(U) \le pd_S(k) = n$.

The Auslander-Buchsbaum Formula

V graded f.g S-mod, $pd_S(V) = n - \mathrm{depth}(V)$.

pf. $b_i^S(V) = \dim_k(\mathrm{Tor}_i^S(V,k)) = \dim_k(H_i(\mathbb{K} \otimes V))$

$\Rightarrow pd_S(V) = \max\{i \mid H_i(\mathbb{K} \otimes V) \ne 0\}$

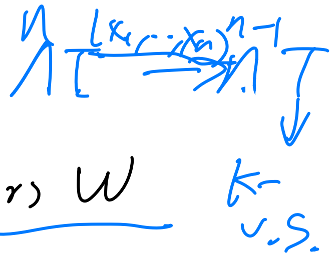
Induction on $j = \mathrm{depth}\ V$.

① $j = 0$, $\mathrm{depth}(V) = 0 \Leftrightarrow$ no $\mathfrak{m}$-regular elements in $\mathfrak{m}$

$\Leftrightarrow$ $\mathfrak{m}$ is associated prime

$\Rightarrow \exists\, u \in V$, s.t. $\mathfrak{m} \cdot u = 0$

$\Rightarrow H_n(\mathbb{K} \otimes V) \ne 0 \Rightarrow pd_S(V) = n$



② Suppose $j-1$ holds

Choose V-regular $r \in \mathfrak{m}$. Set $V/(r)V = W$

$\mathrm{depth}(W) = \mathrm{depth}(V) - 1$

& $\mathrm{depth}(W) = n - pd_S(W)$

WANT. $pd_S(W) = pd_S(V) + 1$

$$0 \to V \xrightarrow{r} V \to W \to 0.$$

$$\Rightarrow \cdots \to H_{i+1}(\mathbb{K} \otimes W) \to H_i(\mathbb{K} \otimes V) \xrightarrow{r} H_i(\mathbb{K} \otimes V) \to H_i(\mathbb{K} \otimes W) \to H_{i-1}(\mathbb{K} \otimes V) \to \cdots.$$

By prop 14.6.

□

Proposition 14.6. *Let W be a finitely generated R-module. We have*

$$(f_1, \dots, f_q)\, \mathrm{H}_i(\mathbf{K}(\mathbf{f}; W)) = 0 \quad \textit{for all} \quad i \ge 0\,.$$

$\Rightarrow r \cdot H_i(\mathbb{K} \otimes V) = 0$ Take $r = x_i$

$\Rightarrow 0 \to \underline{H_i(\mathbb{K} \otimes V)} \to \underline{H_i(\mathbb{K} \otimes W)} \to \underline{H_{i-1}(\mathbb{K} \otimes V)} \to 0$

$\Rightarrow \underline{\max}\{ i \mid H_i(\mathbb{K} \otimes W) \} \neq 0 \}$

$= 1 + \max\{ i \mid H_i(\mathbb{K} \otimes V) \mid \neq 0 \}$

$\Rightarrow pd_S(W) = 1 + pd_S(V)$.

$\square$.

Cor. $pd_S(V) \geq \operatorname{codim}(V) = n - \dim(V)$.

If V is artinian, then $pd_S(V) = n$.

pf. $\operatorname{depth}(V) \leq \dim(V)$.

$\Rightarrow pd_S(V) = n - \operatorname{depth}(V) \geq n - \dim(V)$

$= \operatorname{codim}(V)$

$\square$.

Cor. V graded f.g. S-mod, then $pd_S(V) \geq \operatorname{codim}(V)$

Serre's Theorem

Serre's Theorem 15.10. *The following properties are equivalent.*

(1) *Every graded finitely generated R-module has finite projective dimension.*

(2) $\mathrm{pd}_R(k) < \infty$.

(3) *R is a polynomial ring, that is, $R = S/I$ for some ideal I generated by linear forms.*

See [Matsumura, Theorem 19.2] for a proof of Serre's Theorem.

R/I graded regular ring $\Leftrightarrow$ I is generated by linear forms

X smooth $\hookrightarrow \mathbb{P}^n$

S/I regular $\neq$ X smooth.

Ex. $(S/I)_{m/I}$ is not regular unless S/I is polynomial ring